

Evangelia (Eva) Deliporanidou

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Education

University of Cambridge – DAMTP, PhD in Astrophysics Oct 2023 – Present

- Supervised by Prof. Giulio Del Zanna in the Astrophysics Group, Department of Applied Mathematics and Theoretical Physics (DAMTP). Funded by an STFC (UKRI) Doctoral Studentship.
- PhD Research: Developing physically motivated models of solar and stellar EUV irradiances by combining differential emission measure modelling, atomic physics and proxies of activity. The resulting models enable the reconstruction of stellar spectra for stars where direct EUV observations are unavailable.
- Research collaborations with the University of Cambridge, NASA Goddard Space Flight Centre, NASA's Laboratory for Atmospheric and Space Physics (LASP), the international CHIANTI collaboration, and the ESA F-class mission *SIRIUS*, contributing to the development and validation of stellar XUV irradiance models.
- Contributed to the ongoing development of the CHIANTI (<https://www.chiantidatabase.org>) atomic database, an internationally recognised collaboration honoured with the 2024 NASA Group Achievement Award for its contributions to astrophysics. CHIANTI is universally used in solar physics to analyse spectroscopic observations (over 5,200 citations)
- Extensive scientific programming in Python, IDL and Fortran for spectral synthesis, plasma diagnostics and astrophysical modelling.
- Expected thesis submission: March 2027.

University College London (UCL), MSc in Astrophysics Sept 2022 – Sept 2023

- MSc Thesis: *Towards a 3D Map of the Dust in the Nuclear Stellar Disk of the Milky Way*. Combined infrared observations and statistical modelling to reconstruct the three-dimensional dust distribution within the Galactic Centre, improving our understanding of the Milky Way's central structure. Supervised by Dr Jason Sanders.
- Academic Representative for the Astrophysics MSc Programme (2022-2023)

University of Strathclyde, BSc (Hons) in Physics Sept 2019 – Sept 2022

- Deferred entry into second year
- First Class Honours (80/100)
- BSc Thesis: Atomic Processes for Astrophysical Plasmas - Quadrupole l-changing collisions - Supervised by Prof. Nigel Badnell. Investigated quantum mechanical atomic collision processes governing hydrogen- and helium-like ions, improving theoretical models of astrophysical plasma emission.

Research Interests

Solar and stellar atmospheres • Exoplanet host stars • Atomic physics • Plasma spectroscopy • Extreme-ultraviolet and X-ray spectroscopy • Interstellar Medium (ISM) • Radiative transfer • Stellar magnetic activity

Publications

6 refereed publications (4 first-author), 70+ citations, h-index 3

First-author publications

Modelling Stellar Irradiance III: A predictive model for solar irradiances.

Evangelia Deliporanidou, G. Del Zanna

Monthly Notices of the Royal Astronomical Society (2026)

Developed a physically driven semi-empirical model for predicting solar EUV irradiances, using atomic physics and correlations with proxies of activity such as the F30 cm radio flux.

DOI: [10.1093/mnras/stag412](https://doi.org/10.1093/mnras/stag412)

Modelling Stellar Irradiances – II. Correlations of solar irradiances with proxies of activity along a cycle.

Evangelia Deliporanidou, G. Del Zanna, T. N. Woods, D. Woodraska

Monthly Notices of the Royal Astronomical Society (2025)

Quantified correlations between solar EUV irradiances and multiple proxies of activity, investigating solar cycle variations and evaluating the current solar-irradiance observational data from SDO/EVE and SOHO/CDS.

DOI: [10.1093/mnras/staf812](https://doi.org/10.1093/mnras/staf812)

H, He-like recombination spectra VI: quadrupole l-changing collisions

Evangelia Deliporanidou, N. R. Badnell, P. J. Storey, G. Del Zanna, G. J. Ferland

Monthly Notices of the Royal Astronomical Society (2025)

Calculated quadrupole l-changing collision rates for hydrogen- and helium-like ions, improving theoretical recombination spectra used in astrophysical plasma diagnostics.

DOI: [10.1093/mnras/staf608](https://doi.org/10.1093/mnras/staf608)

Modelling Stellar Irradiances I: The transition regions of FGKM stars

Evangelia Deliporanidou, G. Del Zanna

Monthly Notices of the Royal Astronomical Society (2024)

Developed new transition-region models for FGKM stars using updated ionisation equilibrium calculations to improve stellar EUV spectrum reconstruction.

DOI: [10.1093/mnras/stae2299](https://doi.org/10.1093/mnras/stae2299)

Co-authored publications

CHIANTI—An Atomic Database for Emission Lines—Paper. XVIII. Version 11, Advanced Ionization Equilibrium Models: Density and Charge Transfer Effects

R.P. Dufresne, G. Del Zanna, P. R. Young, K. P. Dere, **E. Deliporanidou**, W. T. Barnes, E. Landi

The Astrophysical Journal - ApJ (2024)

Contributed to Version 11 of the CHIANTI atomic database through the development and validation of improved atomic data for plasma emission modelling.

DOI: [10.3847/1538-4357/ad6765](https://doi.org/10.3847/1538-4357/ad6765)

SIRIUS: stellar and ISM research via in-orbit ultraviolet spectroscopy

Barstow et. al, **Deliporanidou E.**

Proceedings of SPIE, Vol. 14146 (2026)

Contributed to the SIRIUS mission science case, with a particular focus on its extreme-ultraviolet (EUV) science objectives and potential applications.

DOI: [10.1117/12.3104810](https://doi.org/10.1117/12.3104810)

Selected Talks & Presentations

Cool Stars 23 (June 2026) <i>A predictive model for stellar irradiances: From the Sun to Exoplanetary Applications</i> (Poster)	Tokyo, Japan
Royal Astronomical Society Specialist Discussion Meeting (March 2026) <i>A predictive model for solar irradiances</i> (Contributed Talk)	London, UK
National Astronomy Meeting (NAM - July 2025) <i>A predictive model for solar irradiances</i> (Talk)	Durham, UK
European Solar Physics Online Seminars (ESPOS - May 2025) <i>A predictive model for solar irradiances</i> (Invited Seminar)	Online
UK Solar Physics Meeting (UKSP25 - April 2025) <i>A predictive model for solar irradiances</i> (Talk)	Sheffield, UK
European Solar Physics Meeting (ESPM-17 - September 2024) <i>Correlation of solar irradiances with proxies of activity along a cycle</i> (Talk)	Turin, Italy
National Astronomy Meeting (NAM - July 2024) <i>Correlation of solar irradiances with proxies of activity along a cycle</i> (Talk)	Hull, UK
Cool Stars 22 (June 2024) <i>Modelling stellar transition regions of FGKM stars with advanced ionisation equilibrium</i> (Poster)	San Diego, USA

Department of Applied Mathematics and Theoretical Physics, University of Cambridge Cambridge, UK
Departmental research seminars and journal club presentations on solar and stellar irradiance modelling.

Teaching (Academic) Experience

Undergraduates' Supervisor, University of Cambridge Oct 2022 – Present

- Supervised undergraduate students for the Part II course "Classical Dynamics", including material in differential equations, Lagrangian and Hamiltonian and classical mechanics.
- Supervised for 13 Cambridge colleges through the University's intercollegiate supervision system, including, Gonville and Caius, Queen's, Clare College, Magdalene, Jesus, St. Catharine's, Corpus Christi and others.

Exam Invigilator, University of Cambridge Feb 2024 - Present

Supported the smooth running of University examinations by maintaining examination procedures and ensuring a fair assessment environment.

Physics and Maths Tutor, Online Oct 2021 – Oct 2023

Help students learn, understand and comprehend new concepts in Mathematics and Physics, mainly for the IBDP and A-levels. Prepared lessons by studying the student plans, provided specialized notes, reviewed textbooks in detail to provide the appropriately-adjusted-to-each student material.

Research Funding and Awards

Awards & Funding

- **STFC Doctoral Studentship (UK Research and Innovation) - University of Cambridge**
Modelling the emission of the outer solar atmosphere

Conference Travel Awards

- NASA Exoplanet Exploration Program Travel Award - Cool Stars 22, San Diego, USA
- Royal Astronomical Society Travel Grant - UKSP25, Sheffield, UK (2025)

Societies & Activities

- President of the Cambridge University Hellenic Society (2025/26)
- Secretary of the Cambridge University Hellenic Society (2024/25)
- Member of the Royal Astronomical Society (RAS)
- Fundraised over £700 for Alzheimer's Research UK through participation in the Cambridge Half Marathon (2026).
- Fundraised for Churchill college as a student called in the annual fundraising campaign (February 2025 - March 2025)

Academic Service & Outreach

- Volunteer, DAMTP Open Days, University of Cambridge (March 2024, March 2026). Assisted with astronomy outreach activities and interactive science demonstrations for prospective students, school groups and members of the public.
- Volunteer, Institute of Astronomy Open Day, University of Cambridge (March 2025). Assisted with public engagement activities and astronomy demonstrations for visitors and schoolchildren.
- Contributor to the CHIANTI atomic database collaboration.

Languages

Greek (native) - English (fluent) - German (intermediate) - Spanish (beginner)